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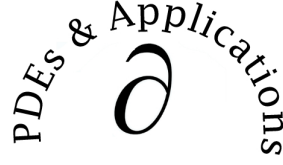
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**Convergence of FV Scheme for a Parabolic System
in Image Processing.**

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Abstract:

We analyze a finite volume scheme for a nonlinear reaction-diffusion system used in image processing. First, we establish the existence of a solution to the scheme. Then, by deriving a series of a priori estimates and employing Kolmogorov's compactness criterion, we prove the convergence of the finite volume solution to the weak solution. Our numerical experiments demonstrate the effectiveness of the proposed model in preserving small details, texture, and fine structures when compared to the modified Perona-Malik nonlinear image selective smoothing equation proposed by Catté, Lions, Morel, and Coll [1].

Keywords: Image Processing, Perona-Malik Equation, Finite Volume Method, Convergence, Nonlinear Reaction-diffusion.

1 Introduction

Our work presents a novel nonlinear reaction-diffusion system that effectively enhances coherence. Taking inspiration from the Perona-Malik diffusion process, we have devised an approach that combines its strengths. The equation we propose, incorporating Neumann boundary conditions, is as follows

$$\left\{ \begin{array}{ll} \frac{\partial u}{\partial t} - \operatorname{div} \left(g(|\nabla G_\sigma * u|) \nabla u \right) + 2\lambda w = 0 & \text{in } \Omega_T := \Omega \times (0, T), \\ \frac{\partial w}{\partial t} = \Delta w - (u_0 - u) & \text{in } \Omega_T := \Omega \times (0, T), \\ \langle g(|\nabla G_\sigma * u|) \nabla u, \nu \rangle = 0, \quad \frac{\partial w}{\partial \nu} = 0 & \text{on } \Sigma_T := \partial\Omega \times (0, T), \\ u(0) = u_0, \quad w(0) = 0 & \text{in } \Omega, \end{array} \right. \quad (1)$$

where $g(s)$ is the decreasing smooth function, $g(0) = 1, 0 < g(s) \rightarrow 0$ for $s \rightarrow \infty$,

$$G_\sigma \in C^\infty(\mathbb{R}^2) \text{ is the smoothing kernel with } \int_{\mathbb{R}^2} G_\sigma(x) dx = 1.$$

Furthermore, $\lim_{\sigma \rightarrow 0} G_\sigma = \delta_x$, where δ_x is the Dirac measure at point x , $u_0 \in L^2(\Omega)$. Also

$$(G_\sigma * u)(x) = \int_{\mathbb{R}^2} G_\sigma(x - \xi) \tilde{u}(\xi) d\xi,$$

where $\tilde{u}$ is an extension of u to $\mathbb{R}^2$ (see [1]) and $G_\sigma(x)$ is the Gaussian filter, given by

$$G_\sigma(x) = \frac{1}{2\pi\sigma^2} \exp\left(-\frac{\|x\|^2}{\sigma^2}\right), \quad x \in \mathbb{R}^2.$$

Moreover, in literature, different diffusion coefficients have been proposed. For example, Perona and Malik [5] provided some examples of g . They proposed diffusion functions of the type

$$g_1(s) = \frac{1}{1 + (s/\alpha)^2} \quad \text{or} \quad g_2(s) = e^{-(s/\alpha)^2}.$$

Note that in our case, we are interested in the first one with $\alpha = 1$.

In this paper, we utilize the finite volume method (referenced as [2]) to derive an approximation scheme for (1). This scheme is well-suited for image processing, as it employs discrete approximations that are piecewise constant, corresponding to control volumes aligned with the pixel structure of the image. Moreover, the scheme allows us to obtain L^2 a-priori estimates for fully discrete solutions. To establish convergence of the approximations towards the weak solution of (1), we rely on Kolmogorov's compactness theorem.

2 Literature Review

Several authors have already investigated the convergence of such studies. In 2001, K. Mikula and N. Ramarosy [4] examined the convergence of the semi-implicit finite volume scheme for the modified Perona-Malik nonlinear image selective smoothing equation (referred to as anisotropic diffusion in image processing) in the sense of Catt, Lions, Morel, and Coll. They established the convergence to the weak solution. Additionally, Z. Kriv [3] studied the explicit finite volume scheme and demonstrated its convergence to the weak solution of the continuous equation. However, this explicit scheme imposes certain restrictions on the scale step size, unlike the semi-implicit scheme.

3 Results and Discussion

We approximate the model as follows: Determine vectors $(u_K^n)_{K \in \mathcal{T}_h}$ and $(w_K^n)_{K \in \mathcal{T}_h}$ for $n \in \{0, \dots, N\}$, such that for all $K \in \mathcal{T}_h$ and $n \in \{0, \dots, N-1\}$

$$m(K) \frac{u_K^{n+1} - u_K^n}{\Delta t} - \sum_{L \in N(K)} g_{KL}^{\sigma, n}(\tilde{u}_h) T_{KL} (u_L^{n+1} - u_K^{n+1}) + 2\lambda m(K) w_K^{n+1} = 0, \quad (2)$$

$$m(K) \frac{w_K^{n+1} - w_K^n}{\Delta t} - \sum_{L \in N(K)} T_{KL} (w_L^{n+1} - w_K^{n+1}) = m(K)(u_K^{n+1} - u_K^0), \quad (3)$$

$$u_K^0 = \frac{1}{m(K)} \int_K u_0(x) dx, \quad g_{KL}^{\sigma, n}(\tilde{u}_h) := g(|\nabla G_\sigma * \tilde{u}_h(x_{KL}, t_n)|), \quad (4)$$

where $\tilde{u}$ is a periodic extension of the discrete image calculated in the n -th scale step. For more details, see [3].

For the purpose of the analysis, we introduce "piecewise constant" functions.

$$u_h(x, t) = u_K^{n+1} \quad \text{and} \quad w_h(x, t) = w_K^{n+1},$$

for all $(x, t) \in K \times (n\Delta t, (n+1)\Delta t]$, with $K \in \mathcal{T}_h$ and $n \in \{0, \dots, N-1\}$.

Our main result is

Theorem 3.1 *Assume that $u_0 \in L^2(\Omega)$. Then the finite volume solution (u_h, w_h) , converges along a subsequence to (u, w) as $h \rightarrow 0$, where (u, w) is a weak solution of (1). The convergence is in the following sense:*

$$(u_h, w_h) \rightarrow (u, w) \text{ strongly in } L^2(\Omega_T) \text{ and a.e. in } \Omega_T,$$

$$(\nabla_h u_h, \nabla_h w_h) \rightarrow (\nabla u, \nabla w) \text{ weakly in } (L^2(\Omega_T))^2.$$

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